

SHIPPER END-TO-END ROLE-STAGE MATRIX

*Multi Tenant Platform Build - canonical shipper flow from ready-for-shipment handoff through delivery in Ghana
Revision 5 - aligned to the live schema/control model, child-exception governance, and the AI-assisted retailer exception architecture*

Purpose. This version supersedes earlier shipper documents. It keeps the explicit ready-for-shipment handoff, makes clear that the shipper only handles the progressed subset, and shows where the shipper is not part of the normal retailer AI loop but does enter the exception path when liability becomes shipper-side.

Role definition

The shipper is the downstream logistics actor. The shipper does not create importer orders, does not upload DVA/card statements, does not set FX, does not operate the importer OCR reconciliation workspace, and is not the normal actor for retailer-facing AI drafted messages. The shipper receives the progressed subset only after the internal ready-for-shipment handoff, monitors inbound parcels using the importer's courier/tracking context, handles UK receipt/intake, booking, dispatch, shipment progression, POD, and delivery in Ghana, and participates in disputes only where the facts point to shipper liability or shipper evidence.

Locked architecture, schema, and control assumptions used here

- Importer creates the order, uploads multiple retailer screenshots, later submits tracking and/or retailer invoice evidence in either order, and operates the OCR reconciliation workspace.
- Supervisor and admin are staff for DVA upload, funding reconciliation, credit application, approval gates, and governance.
- Correct invoice lines may progress while unresolved lines become child exceptions. The parent order is only fully cleared once progressed lines plus resolved children reconcile back to the original parent baseline.
- The explicit ready-for-shipment handoff remains a distinct stage: funded and progressed work becomes shipper-actionable only after internal purchasing/reconciliation has yielded a progressed subset ready for logistics.
- The live schema already includes orders.shipper_id, shipping_quotes, shipping_quote_orders, couriers, order_tracking_submissions, booking_ref, BOL/certificate/commercial invoice/POD fields, Ghana delivery markers, disputes, and shipper_liabilities.
- Retailer exception handling uses a server-side AI drafting service with SOP-aware prompts, but the normal retailer communication loop is operated from the importer/operator side, not by the shipper.
- Where a child exception remains open, the shipper must not absorb that unresolved quantity/value into shipment allocation just because other lines progressed.

Shipper operating boundary

Boundary	Detail
Shipper does	See only their own shipper queue; use courier/tracking context to anticipate inbound retailer parcels; receive and intake the progressed subset; create/update bookings; upload shipment evidence; dispatch; track ETA/SLA; upload POD; mark Ghana delivery; respond to disputes where liable_party = shipper; maintain shipper-side evidence and liability responses.
Shipper does not	Create importer orders; upload DVA statements; fund orders; edit OCR lines; create/delete manual exception lines; select refund/replacement remedy for retailer-side exceptions; bypass the ready-for-shipment handoff; act as the normal retailer AI-communication user; approve

Boundary	Detail
	refunds or accounting postings.
Key control	The shipper only handles the progressed subset. Open child exceptions stay separate and must not be silently mixed into shipment allocation or delivery completion.

Stage 1 - Shipper account presence and visibility scope

Role	Shipper / shipper user
Allowed action	Access the shipper portal and view only the orders/shipments/disputes allocated to that shipper.
Resource(s) / system(s) used	Next.js/Vercel shipper portal; Supabase shipper-user auth and RLS; orders.shipper_id and shipper-scoped read policies.
Data touched	Shipper user auth mapping, shipper identity, shipper queue visibility over orders, shipping quotes, and later shipper-linked disputes/liabilities.
Validation rules	Shipper user must map to the correct shipper and see only that shipper's data.
State transition	Shipper user becomes eligible to monitor shipper-actionable work.
Next actor	Shipper
Exception path	If the shipper mapping is wrong, the user could see the wrong queue or miss their own queue. Fix access before live use.

Stage 2 - See the explicit ready-for-shipment handoff

Role	Shipper / shipper user
Allowed action	View only orders/line subsets that have already been funded, reconciled, and explicitly marked ready for shipment by the upstream internal process.
Resource(s) / system(s) used	Shipper portal; Supabase orders, shipping_quote_orders, progress views/read models.
Data touched	orders.shipper_id, funded/progressed readiness indicators, shipment allocation readiness, child-exception visibility flags.
Validation rules	The shipper must not treat funded-but-not-progressed work as actionable. The handoff is explicit: only the progressed subset enters the shipper lane.
State transition	Work becomes shipper-actionable without waiting for all child exceptions to close, provided the progressed subset is clearly defined.
Next actor	Shipper
Exception path	If unresolved child quantity/value is mixed into the actionable subset, the handoff is wrong and must be corrected upstream.

Stage 3 - Use courier/tracking context to anticipate inbound retailer parcels

Role	Shipper / shipper user
Allowed action	Read the importer-submitted courier, tracking reference, and tracking date to know which courier site/template to use and when to expect parcel arrival at the UK side.

Resource(s) / system(s) used	Shipper portal; Supabase order_tracking_submissions; couriers lookup with tracking URL template.
Data touched	order_tracking_submissions.courier_id, tracking_ref, tracking_date, optional screenshot, active vs superseded submission history.
Validation rules	Courier is mandatory. The shipper reads this context but does not create the importer's tracking submission in the normal flow.
State transition	Shipper gains parcel-anticipation visibility before or after invoice timing.
Next actor	Shipper
Exception path	Wrong courier, wrong tracking ref, or stale superseded submission makes the ETA unreliable and should be corrected before relying on it.

Stage 4 - Receive and intake the progressed subset at the UK side

Role	Shipper / shipper user
Allowed action	Receive physical goods/parcels at the UK side and record what arrived, in what condition, and whether the progressed subset matches expectation.
Resource(s) / system(s) used	Shipper portal; evidence uploads; shipment/intake screens; Supabase receipt-related status fields and notes.
Data touched	Receipt status, intake notes, condition notes/images, partial/damaged/not-arrived evidence.
Validation rules	Only the progressed subset should be handled. Intake evidence must preserve auditability and not rewrite upstream truth.
State transition	Goods move from expected inbound to received / partially received / damaged / missing at the UK side.
Next actor	Shipper and internal staff
Exception path	If goods do not arrive or arrive damaged/partial, the exception path begins immediately and later liability may fall on retailer, shipper, or unknown depending on the facts.

Stage 5 - Create or update the shipment booking

Role	Shipper / shipper user
Allowed action	Create the shipping booking/consignment and attach the progressed orders/lines that are actually moving to Ghana.
Resource(s) / system(s) used	Shipper portal; Supabase shipping_quotes and shipping_quote_orders; courier lookup; internal apportionment records.
Data touched	shipping_quotes.quote_gbp_total, booking_ref, courier_id, status, shipping_quote_orders.order_id, apportionment_pct, apportioned_shipping_gbp.
Validation rules	Only progressed/eligible quantity/value should be attached. Open child exceptions must not be smuggled into the shipment just because they belong to the same parent order.
State transition	A booking exists and the shipment plan becomes concrete and auditable.
Next actor	Shipper

Exception path	Wrong order allocation, wrong courier, or contamination by unresolved child exceptions must be corrected before dispatch.
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Stage 6 - Upload shipment evidence and dispatch details

Role	Shipper / shipper user
Allowed action	Upload booking and dispatch artefacts, then move the shipment from draft/confirmed into dispatched when it really leaves the UK side.
Resource(s) / system(s) used	Shipper portal; Supabase shipping_quotes; storage; image compression where images are uploaded.
Data touched	booking_ref, bol_url, cert_of_shipment_url, commercial_invoice_url, dispatched_at, status, estimated_ghana_arrival_at.
Validation rules	Evidence must be attached to the correct shipment. Dispatch must not be stamped before actual dispatch.
State transition	Shipment becomes visible as in transit to Ghana.
Next actor	Shipper and importer (visibility); supervisor monitors
Exception path	Weak or missing evidence damages later dispute defence. False dispatch stamping corrupts ETA/SLA monitoring.

Stage 7 - In-transit monitoring and SLA discipline

Role	Shipper / shipper user
Allowed action	Maintain shipment progress while the consignment is en route and keep ETA/SLA information current.
Resource(s) / system(s) used	Shipper portal; Supabase shipping_quotes status and ETA fields; importer-facing tracking views consume the result.
Data touched	shipping_quotes.status, estimated_ghana_arrival_at, sla_dispatch_target_date, sla_breach_flag, sla_breach_reason.
Validation rules	Transit status and SLA flags should reflect reality, not cosmetic updates.
State transition	Shipment stays in dispatched/in-transit state until delivery or a logistics exception emerges.
Next actor	Shipper
Exception path	If the shipment stalls or breaches SLA, the evidence chain and later liability analysis become more important, not less.

Stage 8 - Ghana delivery and proof of delivery

Role	Shipper / shipper user
Allowed action	Complete Ghana delivery and upload/maintain proof that delivery occurred.
Resource(s) / system(s) used	Shipper portal; Supabase shipping_quotes; evidence storage.
Data touched	pod_ghana_url, ghana_delivered_at, delivery status, final proof/evidence notes.
Validation rules	Delivery should not be marked complete without POD. POD is a core downstream artifact because the

	importer confirms receipt or disputes from this state.
State transition	Shipment reaches delivered state and triggers importer receipt confirmation/discrepancy options.
Next actor	Importer
Exception path	If POD is missing or weak, Ghana delivery may be disputed later even if the shipper believes delivery occurred.

Stage 9 - Support discrepancy handling where the issue becomes shipper-side

Role	Shipper / shipper user
Allowed action	Respond to disputes only where the facts implicate the shipper - for example UK-side damage, in-transit loss, Ghana delivery failure, or weak POD/booking evidence.
Resource(s) / system(s) used	Shipper portal; Supabase disputes, dispute_lines, dispute_messages, dispute_notes, dispute_images, shipper_liabilities.
Data touched	liable_party = shipper, shipper responses, evidence uploads, notes, settlement method fields.
Validation rules	The shipper is not the normal retailer AI loop actor. The shipper enters the exception path when the issue becomes shipper liability or requires shipper evidence/response.
State transition	Dispute moves toward accepted, disputed, partial, offset, cash refund, write-off, or further investigation.
Next actor	Supervisor/Admin for final governance and accounting outcome
Exception path	If the shipper response is weak or evidence is poor, staff may resolve conservatively and book liabilities or write-offs with weaker operational support.

Stage 10 - Historical visibility and controlled closure

Role	Shipper / shipper user
Allowed action	Review completed shipments, historical POD/evidence, and any liabilities or closed disputes tied to that shipper.
Resource(s) / system(s) used	Shipper portal history pages; Supabase historical reads over shipping_quotes, disputes, shipper_liabilities.
Data touched	Closed booking records, POD references, dispute outcomes, liability settlements.
Validation rules	Historical truth should not be silently rewritten. Closure must remain traceable from booking through POD and any later dispute resolution.
State transition	Shipment leaves active work and becomes historical unless reopened through a controlled exception path.
Next actor	No active actor in the happy path
Exception path	If a post-delivery issue emerges later, staff reopen through controlled dispute handling rather than editing history directly.

Shipper happy-path summary

- Shipper user is onboarded and sees only their own shipper queue.
- Only funded and progressed work enters the shipper lane. The ready-for-shipment handoff is explicit.

- Importer-submitted courier/tracking context helps the shipper anticipate inbound parcels.
- Shipper receives the progressed subset, creates the booking, uploads dispatch evidence, monitors ETA/SLA, and completes POD/delivery in Ghana.
- Importer then confirms receipt or raises a discrepancy from the delivered state.

Shipper exception-path summary

- Wrong courier/tracking context makes parcel monitoring unreliable and should be corrected before reliance.
- Partial receipt or damage at the UK side can start the exception path early, even before Ghana delivery.
- Already-progressed lines can still move while unresolved child exceptions stay open separately.
- The shipper is not the normal retailer AI actor. Retailer communication remains on the importer/operator side unless the issue becomes shipper liability.
- If goods are damaged, missing, or not properly delivered in Ghana, the shipper may become a liable party and shipper_liabilities becomes the formal recovery lane.

Order-to-Ghana validation checks for the shipper lane

Simulation A - Tracking first, then invoice, then shipment

Importer submits courier + tracking first, giving the shipper early visibility of inbound parcels. Invoice arrives later, importer reconciles and progresses the correct lines, then the shipper receives only that progressed subset, books shipment, dispatches, and uploads POD in Ghana. Validation result: passes.

Simulation B - Invoice first, then tracking, then shipment

Invoice exists before tracking. Importer reconciles and progresses correct lines first. Tracking arrives later. The shipper still uses the later courier context for parcel awareness while proceeding with booking and shipment once the progressed subset is operationally ready. Validation result: passes.

Simulation C - Progressed subset ships while refund child waits for approval

Four clean lines progress and become shipper-actionable. One child exception is set to refund and sits in refund_pending_approval, blocked from retailer refund progression. The shipper still handles only the progressed subset and must not absorb the blocked child into shipment allocation. Validation result: passes and confirms the explicit handoff/control boundary.

Simulation D - Ghana-side damage with shipper liability

Shipment is delivered in Ghana but damage is alleged. The issue becomes shipper-side, the shipper responds with evidence and a liability stance, and recovery/settlement is tracked separately from the already-progressed shipment history. Validation result: passes.

Shipper resource map by node

Node	Primary resource(s)
Shipper queue and portal	Next.js/Vercel shipper portal; Supabase shipper-scoped RLS/data access.
Ready-for-shipment handoff	Orders/progress views; shipping_quote_orders; explicit progressed-subset visibility.
Tracking awareness	order_tracking_submissions; couriers lookup and tracking URL template.
Booking and dispatch	shipping_quotes; booking_ref; BOL/certificate/commercial invoice evidence URLs.
Transit / ETA / SLA	shipping_quotes ETA/SLA fields; importer-facing tracking consumes the result.

Node	Primary resource(s)
Ghana delivery / POD	shipping_quotes POD and delivery fields; storage for proof artifacts.
Shipper-side dispute / liability	disputes, dispute_messages/notes/images, shipper_liabilities.
Not the normal retailer AI loop	Retailer SOP + AI drafting service exist, but normal retailer comms sit with importer/operator unless the matter becomes shipper liability or evidence response.

Internal working document - canonical shipper role matrix for the Multi Tenant Platform Build (Revision 5)